EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Oxidation of alcohol by MnO_2

1. Which one of the following alcohols are oxidised by MnO_2?

(A) $C_6H_5-CH_2-CH_2-OH$

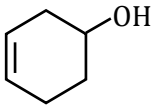
(B) $CH_2=CH-CH_2-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$

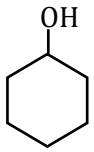
(C) $CH_3-CH=CH-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$

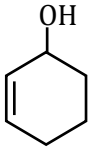
(D) $CH_3-CH_2-CH_2-OH$

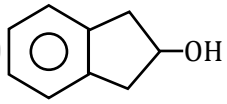
COX007

2. Which of the following can be oxidised by MnO_2:

(A) 

(B) 

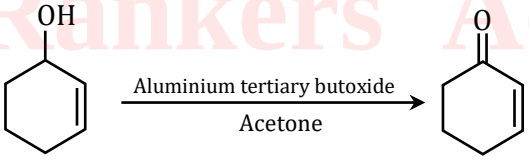
(C) 

(D) 

COX009

Oxidation of alcohol by Oppenaur's Oxidation

3. The given reaction



is knows as :

(A) Kolbe reaction
(B) Tischenko reaction
(C) MPV reaction
(D) Oppeneur oxidation

COX001

Oxidation of diol by HIO_4 and $Pb(OAc)_4$

4. In the given reaction :

$$CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-\underset{\displaystyle OH}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-CH_3 \xrightarrow{HIO_4} (a) + (b)$$

(a) and (b) respectively be :

(A) CH_3CHO and CH_3CHO
(B) CH_3COCH_3 and CH_3CHO
(C) CH_3COCH_3 and CH_3COCH_3
(D) CH_3COOH and CH_3COCH_3

COX002

Oxidation of carbonyl compound; Tollen's Reagent

5. $CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-H$ and $CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-CH_3$ is differentiated by

(A) Tollen's reagent
(B) Lucas test
(C) Iodoform test
(D) $NaHSO_3$ test

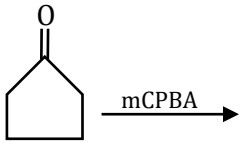
COX003

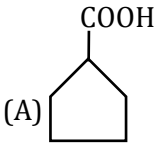
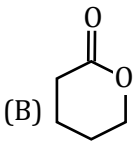
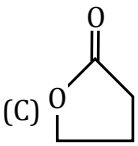
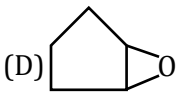
Rankers Academy JEE
For More Join: @IITJEE_Advanced

6. Which will give silver mirror test with Tollens reagent :
(A) C_6H_5CHO (B) $CH_3–CHO$ (C) HCOOH (D) All of these

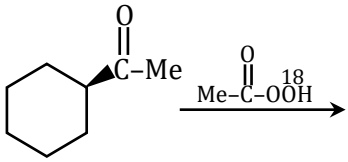
COX005

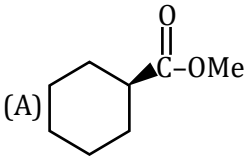
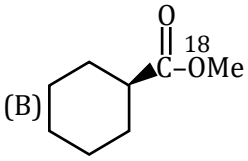
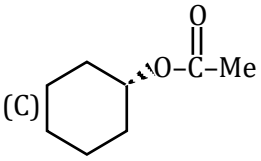
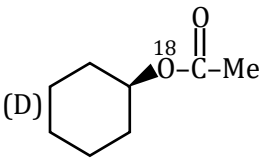
Oxidation of carbonyl compound; Baeyer Villiger oxidation

7.  Major product is :

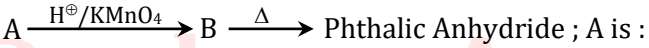





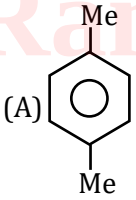
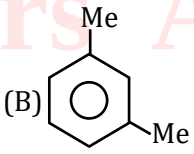
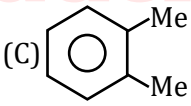
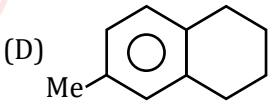
COX008

8.  Major product is :

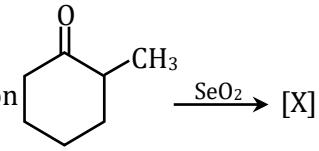
COX010

9. 

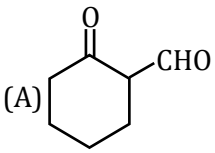
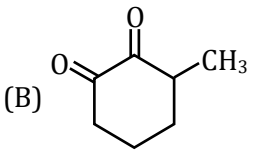
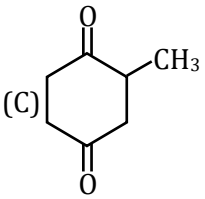
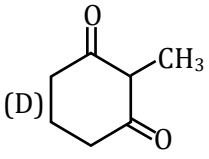





COX011

SeO_2

10. In the given reaction  [X]

[X] will be :

COX004

Rankers Academy JEE
For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

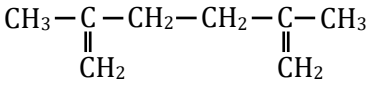
Oxidation of alcohol by mild oxidising agent [PCC, PDC, NBS, TsCl + DMSO + $NaHCO_3$, Sharett Reagent (PCC in CH_2Cl_2), Collin Reagent]

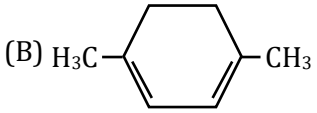
1. Propan-1-ol and Propan-2-ol can be best distinguished by
(A) oxidation with alkaline $KMnO_4$ followed by reaction with H_2O
(B) oxidation with PCC followed by reaction with Tollen's reagent
(C) oxidation by heating with copper followed by reaction with iodoform test
(D) reaction with cone. H_2SO_4 followed by reaction with Fehling solution.

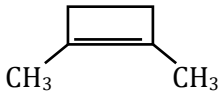
COX079

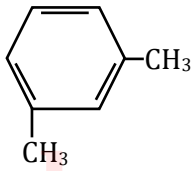
Ozonolysis of alkenes (Reductive and oxidative)

2. An organic compound on reaction with O_3 followed by Zn and H_2O gives $CH_3COCH_2CH_2COCH_3$. The structure(s) is/are

(A) 

(B) 

(C) 

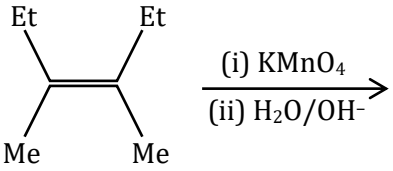
(D) 

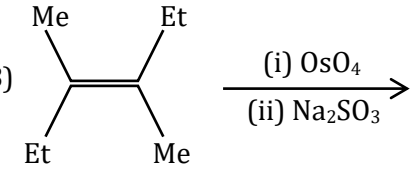
Rankers Academy JEE

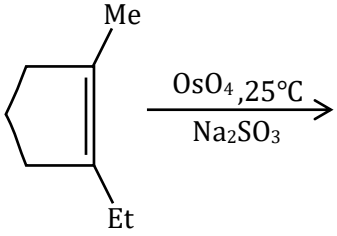
COX076

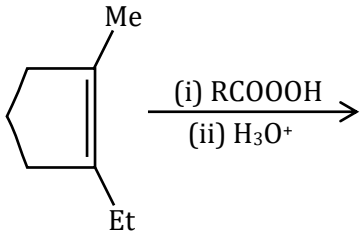
Oxidation of Alkenes (Baeyer's Reagent, OsO_4, Hot $KMnO_4$, Acidic $KMnO_4$, Peroxy Acid followed by hydrolysis, Ag/O_2 followed by hydrolysis)

3. Which of the following give syn addition ?

(A) 

(B) 

(C) 

(D) 

COX077

4. On oxidation, an unknown alkane C_8H_{16} gives a mixture of propanoic acid and pentanoic acid. The unknown alkene can be
(A) cis-3-octene
(B) trans-3-octene
(C) cis-4-octene
(D) trans-4-octene

COX078

For More Join: @IITJEE_Advanced

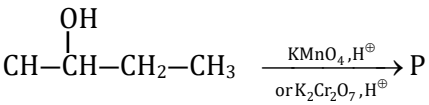
EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Oxidation of alcohol by Strong oxidising agent ($KMnO_4$, $K_2Cr_2O_7$, CrO_3, H_2SO_4)

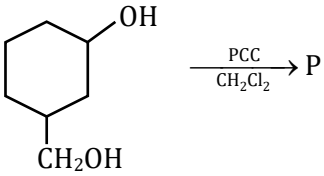
1. What is the total number of π–bond in product P ?



COX085

Oxidation of alcohol by MnO_2

2. What is the total number of oxygen atom product P ?



COX086

Oxidation of carbonyl compound; Tollen's Reagent

3. How many of the following compounds will give Fehling test positive.
HCHO, PhCHO, MeCHO, CH_3COCH_3 ?

COX089

Rankers Academy JEE

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Oxidation of the Hydrocarbon

4. How many litre of Oxygen required for complete combustion of 6.0 g ethane at NTP ?

COX107

5. 30 mL mixture of ethylene and Butylene is burnt in presence of oxygen then 150 mL of oxygen is required, what is the of volume of Ethylene in mixture.

COX108

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

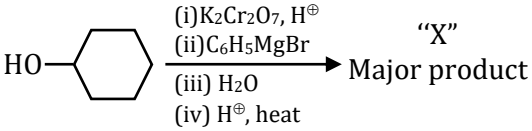
1. $CH_3CH_2-OH \xrightarrow[(ii)\ KMnO_4\ (iii)NaOH,\ CaO,\Delta]{(i)\ Jone's\ Reagent} P$

Consider the above reaction sequence and identify the major product P. **[JEE (Main) 2024]**

(1) Methane (2) Methanal

(3) Methoxymethane (4) Methanoic acid

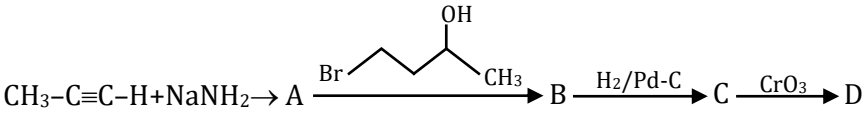
COX111

2. In the given reaction 

the number of sp^2 hybridised carbon (s) in compound 'X' is____. **[JEE (Main) 2022]**

COX050

3. In the following sequence of reactions, the final product D is : **[JEE (Main) 2021]**



(1) $CH_3-CH_2-CH_2-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-H$ (2) CH_3-CH=CH-CH_2-CH_2-CH_2-COOH

(3) H_3C-CH=CH-CH(OH)-CH_2-CH_2-CH_3 (4) $CH_3-CH_2-CH_2-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$

COX043

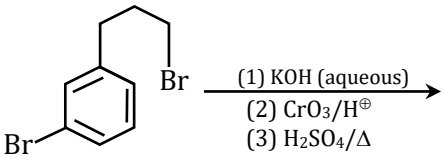
4. Which one of the following reactions will not form acetaldehyde? **[JEE (Main) 2021]**

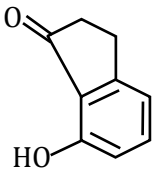
(1) $CH_3CH_2OH \xrightarrow[573K]{Cu}$ (2) $CH_3CN \xrightarrow[(ii)\ H_2O]{(i)\ DIBAL-H}$

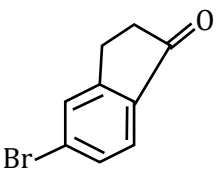
(3) $CH_2=CH_2 + O_2 \xrightarrow[(ii)\ H_2O]{(i)\ pd\ (ii)\ /\ Cu(ii)}$ (4) $CH_3CH_2OH \xrightarrow{CrO_3-H_2SO_4}$

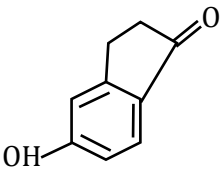
COX040

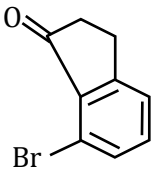
5. The major product of the following reaction is: **[JEE (Main) 2019]**



(1) 

(2) 

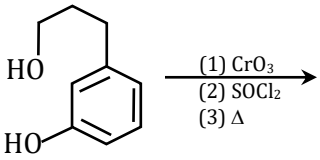
(3) 

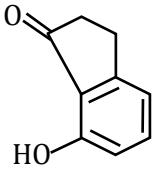
(4) 

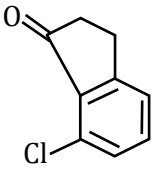
COX036

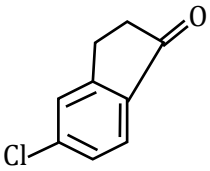
Rankers' Academy JEE
For More Join: @IITJEE_Advanced

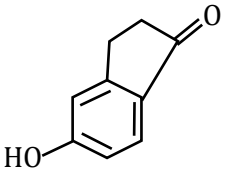
6. The major product of the following reaction **[JEE (Main) 2019]**



(1) 

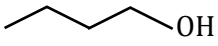
(2) 

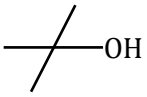
(3) 

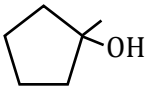
(4) 

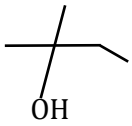
COX037

7. Which of the following will change the colour of acidic dichromate solution. **[JEE (Main) 2013]**

(1) 

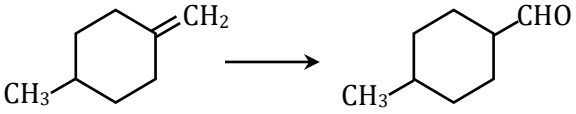
(2) 

(3) 

(4) 

COX032

8. Which of the following reagents/ reactions will convert 'A' to 'B'? **[JEE (Main) 2022]**



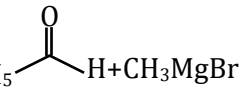
(A) (B)

Rankers Academy JEE

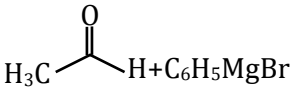
(1) PCC oxidation
(2) Ozonolysis
(3) BH_3, H_2O_2 / ^-OH followed by PCC oxidation
(4) HBr, hydrolysis followed by oxidation by $K_2Cr_2O_7$

COX044

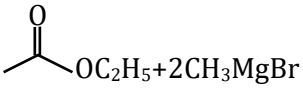
9. Which of the following conditions or reaction sequence will NOT give acetophenone as the major product ? **[JEE (Main) 2022]**

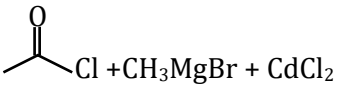
(1) (a) 

(b) $Na_2Cr_2O_7$, $H^{\oplus}$

(2) (a) 

(b) PCC, DCM

(3) 

(4) 

COX049

10. Hex-4-ene-2-ol on treatment with PCC gives 'A'. 'A' on reaction with sodium hypoiodite gives 'B', which on further heating with soda lime gives 'C'. The compound 'C' is **[JEE (Main) 2022]**

(1) 2- pentene (2) proponaldehyde (3) 2-butene (4) 4-methylpent-2-ene

COX048

For More Join: @IITJEE_Advanced

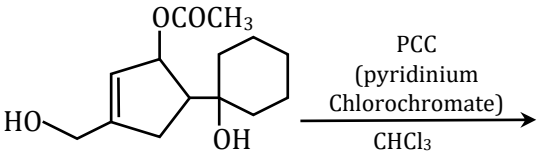
11. The most suitable reagent for the conversion of $R–CH_2–OH \rightarrow R–CHO$ is :- **[JEE (Main) 2014]**

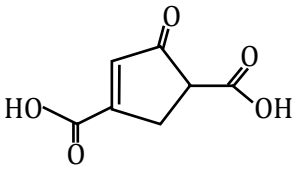
(1) CrO_3 (2) PCC (Pyridinium chlorochromate)

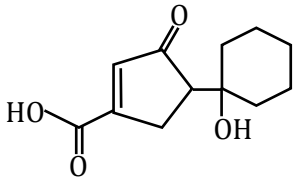
(3) $KMnO_4$ (4) $K_2Cr_2O_7$

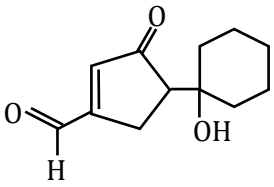
COX034

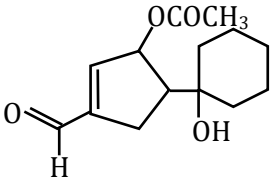
12. The major product formed in the following reaction is :- **[JEE (Main) 2018]**



(1) 

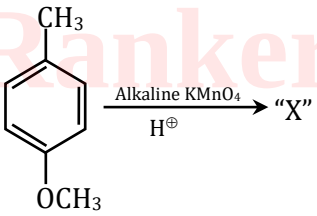
(2) 

(3) 

(4) 

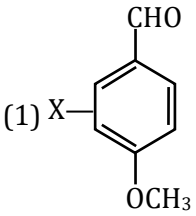
COX035

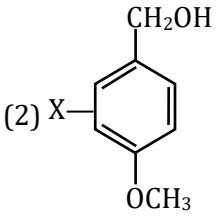
Rankers Academy JEE

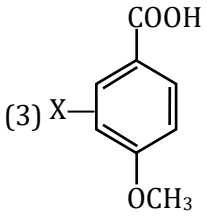
13. 

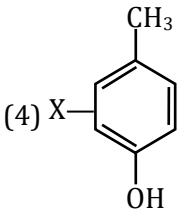
[JEE (Main) 2021]

Considering the above chemical reaction, identify the product "X" :

(1) 

(2) 

(3) 

(4) 

COX042

14. Given below are two statements : **[JEE (Main) 2021]**

Statement-I : 2-methylbutane on oxidation with $KMnO_4$ gives 2-methylbutan-2-ol.

Statement-II : n-alkanes can be easily oxidised to corresponding alcohols with $KMnO_4$.

Choose the correct option :

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

COX041

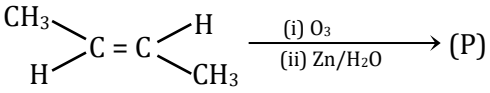
For More Join: @IITJEE_Advanced

15. Which of the following reagent is used for the following reaction ? **[JEE (Main) 2021]**

$CH_3CH_2CH_3 \xrightarrow{?} CH_3CH_2CHO$

(1) Manganese acetate
(2) Copper at high temperature and pressure
(3) Molybdenum oxide
(4) Potassium permanganate

COX039

16. 

Consider the given reaction. The total number of oxygen atoms present per molecule of the product (P) is___. **[JEE (Main) 2024]**

COX109

17. The major product of the following reaction is P.

$CH_3C \equiv C - CH_3 \xrightarrow[273\,K]{\substack{(i)Na/liq.NH_3 \\ (ii)dil.KMnO_4}} 'P'$

Number of oxygen atoms present in product 'P' is _____ (nearest integer). **[JEE (Main) 2024]**

COX110

18. 'A' and 'B' respectively are: **[JEE (Main) 2022]**

A $\xrightarrow[(2)\,Zn-H_2O]{(1)\,O_3}$ Ethane-1,2-dicarbaldehyde + Glyoxal/Oxaldehyde

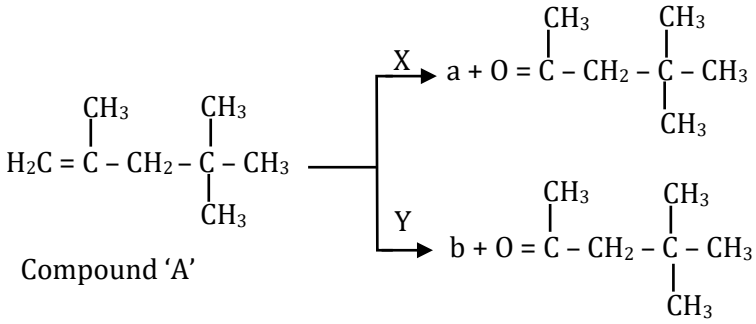
B $\xrightarrow[(2)\,Zn-H_2O]{(1)\,O_3}$ 5-oxohexanal

Rankers Academy JEE

(1) 1-methylcyclohex-1, 3-diene & cyclopentene
(2) Cyclohex-1, 3-diene & cyclopentene
(3) 1-methylcyclohex-1,4-diene & 1-methylcyclopent-1-ene
(4) Cyclohex-1,3-diene & 1-methylcyclopent-1-ene

COX045

19. A compound 'A' on reaction with 'X' and 'Y' produces the same major product but different by product 'a' and 'b'. Oxidation of 'a' gives a substance produced by ants. **[JEE (Main) 2022]**

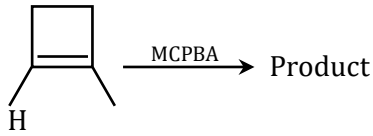


'X' and 'Y' respectively are :

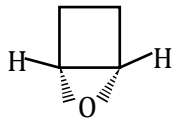
(1) $KMnO_4/H^{\oplus}$ and dil. $KMnO_4$, 273 K
(2) $KMnO_4$,(dilute), 273 K and $KMnO_4/H^{\oplus}$
(3) $KMnO_4/H^{\oplus}$ and O_3, H_2O/Zn
(4) O_3, H_2O/Zn and $KMnO_4/H^{\oplus}$

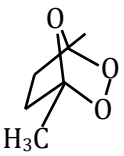
COX046

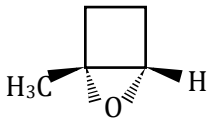
For More Join: @IITJEE_Advanced

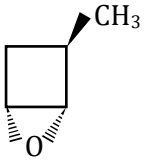
20. 

[JEE (Main) 2013]

(1) 

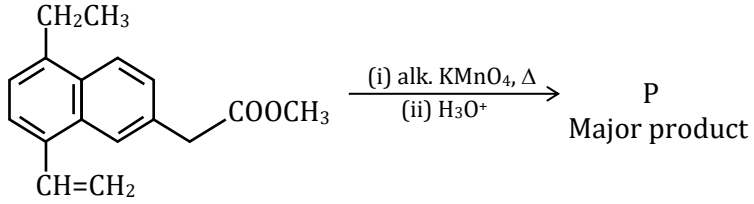
(2) 

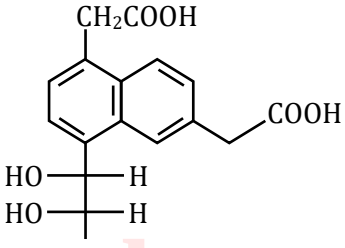
(3) 

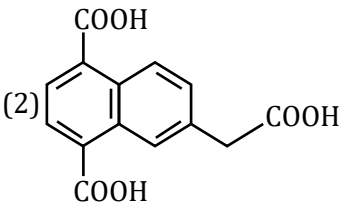
(4) 

COX033

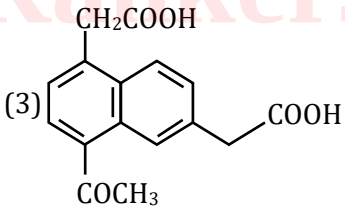
21. The major product 'P' formed in the given reaction is **[JEE (Main) 2023]**

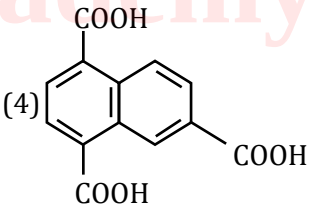


(1) 

(2) 

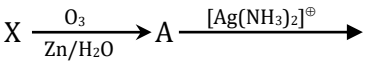
Rankers Academy JEE

(3) 

(4) 

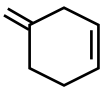
COX090

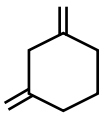
22. An unsaturated hydrocarbon X absorbs two hydrogen molecules on catalytic hydrogenattion, and also gives following reaction : **[JEE (Main) 2020]**

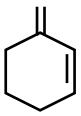


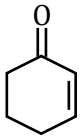
B (3-oxo-hexanedicarboxylic acid)

X will be :-

(1) 

(2) 

(3) 

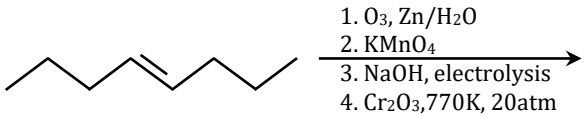
(4) 

COX038

For More Join: @IITJEE_Advanced

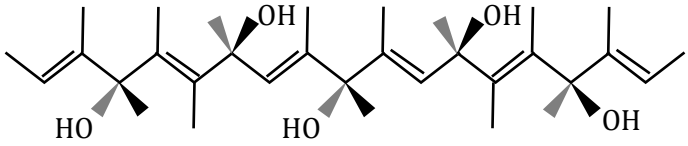
EXERCISE - JEE (Advanced) PYQ

1. The number of $-CH_2-$ (methylene) groups in the product formed from the following reaction sequence is _____. **[JEE (Advanced) 2022]**



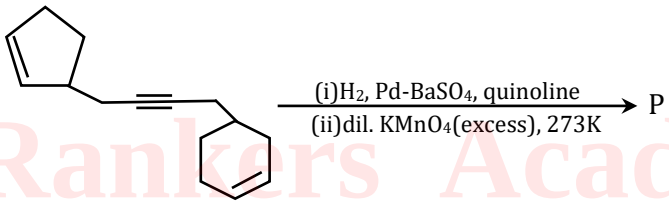
COX074

2. The total number of chiral molecules formed from one molecule of P on complete ozonolysis (O_3, Zn/H_2O) is _____. **[JEE (Advanced) 2022]**



COX075

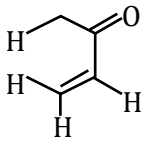
3. Total number of hydroxyl groups present in a molecule of the major product P is **[JEE (Advanced) 2019]**

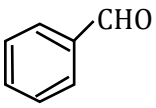


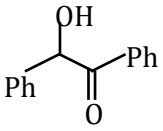
Rankers Academy JEE

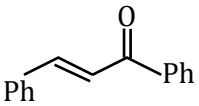
COX073

4. Positive Tollen's test is observed for **[JEE (Advanced) 2016]**

(A) 

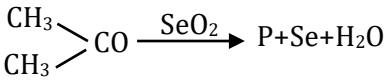
(B) 

(C) 

(D) 

COX072

5. In the reaction, P is **[IIT 1995]**



(A) CH_3COCHO (B) CH_3COOCH_3 (C) CH_3COCH_2OH (D) None

COX071

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	D	B	A	D	B	D	C	B

EXERCISE (O-2)

Que.	1	2	3	4
Ans.	BC	ABC	ABC	AB

EXERCISE (S)

Que.	1	2	3	4	5
Ans.	1	2	2	15.68	10.00

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	8	4	4	2	4	1	3	3	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	3	3	3	1	2	4	4	3
Que.	21	22								
Ans.	4	1								

Bankers Academy JEE

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5
Ans.	0	2	6.00	ABC	A

For More Join: @IITJEE_Advanced

Important Notes

Rankers Academy JEE

For More Join: @IITJEE_Advanced

Important Notes

Rankers Academy JEE

For More Join: @IITJEE_Advanced